

Project 1

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Summary

In this project there will be analyzes about annuity contracts. Consider a directly starting annuity which pays 1 unit of money at the end of each year for as long as the insured is alive between 65 and 90 years old. The first payment is if the insured individual reach the age of 66 years old. In seperate files will provide with lifetable and intrerest for specific time periods. The project will simulate contracts, estimate parametrics, and compare the theortical values and the simulated values with each other.

Simulate 5 000 i.i.d. for a today 65 year old individual.

To be able to calculate the theortical expected value for both in total and for payments in diffrent time periods, is the following formula applied:

$$\mathbb{E}[L_{\nu,\tau}^x] = \sum_{t \geq \nu+1-x} d(t)b_t \mathbb{P}(T_x > t).$$

By using the formula above, the theortical expected value for total payements will be equal to following:

$$\mathbb{E}[L_{65,25}^{65}] = \sum_{t \geq 1} d(t)b_t \mathbb{P}(T_x > t) = 16.22056$$

Moving on to the expected value for for payments in diffrent time periods, the periods will be a age span for 5 years, so in general 65-70,70-75,...,85-90. The things that will change when applying the age spans is the age when the first payment is made ν and τ the end of the time span.

In “Table 1” the theoretical expected value is listed with both the expected value for payments in diffrent periods and the total payment. From “Table 1” a conclusion can be made that for each time period the payment drops.

Table 1: Theortical expected value for both in total and for payments in diffrent time periods

	Expected value
65-70	4.697834
70-75	4.006792
75-80	3.255446
80-85	2.497675
85-90	1.762808
65-90	16.220555

Table 2: Simulated and theortical expected value for both in total and for payments in diffrent time periods

	Simulated value	Theoretical value
65-70	4.697615	4.697834
70-75	4.902785	4.006792
75-80	3.989582	3.255446
80-85	3.135998	2.497675
85-90	2.182447	1.762808
65-90	16.232693	16.220555

For the simulation the same forumla as above is used, the main thing that is diffrent from the theoretical and simulated expected value is that instead of using the factor $\mathbb{P}(T_x > t)$, the factor qx will be used to simulate how long a single individual is alive from a starting age x . This will be applied as following:

if an x -year old individual has the stament that will be definid as $\tau := qx < t \sim U(0, 1)$ the person lives one year, this will be repeated until the statment is false. This will generate how long a single individual will live. The time that the person is alive will then be applied as τ and the factor $\mathbb{P}(T_x > t)$ will be removed becuase of the knowlegde of the individuals lifetime. Conclusion, the formula that will be using for the simulated expected value will be the following (note that τ is a stochastic variabel in this case):

$$\mathbb{E}[L_{\nu, \tau}^x] = \sum_{t \geq \nu+1-x} d(t)b_t.$$

The diffrence between the simulated and theortical expected value for both in total and for payments in diffrent time periods can be seen in “Table 2”. The calculation in diffrent time periods for the simulated can be hard to analyze becuase of the factor that kills and lets and individual to live changes between each simulation, but the same pattern can be seen, the payment drops in each period. The conclusion from analyzing “Table 2” is that the diffrence between the simulated and theortical expected value for in total is very litle, the diffrence is almost not noticeble.

Moving on to the variance of the simulation. By using the diffrence between 5000 simulated contracts expected value and one individuals simulated expected value power to two, the variance can be obtained and which is listed in “Table 3”, the theortical variance is also listed in the same table, which makes it easy to see that even the simulated variance is close to the theortical. For calculating the theortical variance following formula is used:

$$Var(L_{\nu, \tau}^x) = \sum_{t \geq \nu+1-x} d(t)^2 b_t^2 \mathbb{P}(T_x > t)(1 - \mathbb{P}(T_x > t)) + 2 \sum_{t \geq \nu+1-x} \sum_{u > t} d(t)dt(u)b_t b_u \mathbb{P}(T_x > u)(1 - \mathbb{P}(T_x > t))$$

Table 3: Simulated and theortical variance

Simulated variance	Theoretical variance
18.89928	19.09773

If looking in to the quantiles of every individuals payment, which is listet bellow, then it’s easy to see that 95%-quantile has an payment on 18.84639 same for 75%-,50%-quantile, the expection is for the 25%-quantile where the payment is 14.82707

##	0%	25%	50%	75%	100%
##	0.00000	14.82707	18.84639	18.84639	18.84639

Further on the dependence of future payments can be discussed as following: For an individual to get an payment at age 80, the person need be alive at age 79, and that has generated payments up until age 79. So the dependence of future payments is connected with if the person is alive a year before the wanted payment, this can be discussed as if the payments starts at 66, then if a today 65 year old person will receive a payment at age 70 the person has to live through age 67,68,... every year up until age 70, this can then be discussed as the chance of a today x-year old individual live t more years (Survival function). So in order to get and payment t years into the future the probability to die for each of the remaining years until the wanted age will be a factor.

Simulate 5 000 independent annuities, where stress on different factors are applied.

In this case a relative decreasing trend of 1% in the mortality rates is introduced, by using the cumulative intensity function $A(t)$, then

when introducing a relative decreasing trend of 1 % in the mortality rates and compare the expected cash-flow and the variance of the total payments made with the results from (i), i.e. if we let $A(t)$ denote the cumulative intensity function, the following formula can be rewritten with an applied stress to the mortality:

$$P(T_x > t) = \prod_{i=1}^t P(T_x > i | T_x > i - 1)$$

This formula can now be rewritten with help of $A(t)$ with an 1% stress applied:

$$P(T_x > t)_{trend} = \prod_{i=1}^t (A(t+x) - A(t+x-1)) \cdot (1 - 0.01)^t$$

Comparing the expected cash-flow and the variance of total payments between using the first formula (no stress applied on the mortality), and the second one where stress to the mortality is applied, the conclusion from looking at “Table 4” is that an decreasing mortality rate generate an higher cash-flow. This isn’t strange at all, if applying an lower death probability, the individual will live longer which makes the expected cash-flow to increase. In “Table 5” it shows that the theoretical variance is also higher but in this case the simulated was lower, but if the simulation is repeated the simulated variance of L_{trend} will be higher than the variance of L most of the times, but in this case it didn’t happen.

Table 4: Simulated and theoretical expected value for total payments

	Simulated value	Theoretical value
$E[L_{trend}]$	16.44357	16.54609
$E[L]$	16.38347	16.22056

Table 5: Simulated and theoretical variance

	Simulated variance	Theoretical variance
$Var(L_{trend})$	18.36160	19.59565
$Var(L)$	18.71289	19.09773

Further on, an relative decrease of $s\%$ across all mortality rates can also be introduced, and s will be chosen so that $E[L_{flat}] = E[L_{spot}]$. This stress as the above can be expressed as following:

$$P(T_x > t)_{flat} = \prod_{i=1}^t (A(t+x) - A(t+x-1)) \cdot (1 - s/100)$$

With an simple “trial and error”-metod, the value of s was calculated to 9 – 10, but in further calculation $s = 10$. From analyzing “Table 6” where both the simulated and theortical expected value is presented, the same conclusion as for the expected value of L_{trend} , this scenerio will generate that the expected value is higher, both in simulation and theortical, and as before an decreasing mortality rate overall will generate that the individuals live longer which equals to higher payments therefore an higher expected value. The variance in this case by applying the stress is decreased, which is showed in “Table 7”.

Table 6: Simulated and theortical expected value for total payments

	Simulated value	Theortical value
$E[L_{flat}]$	16.43799	16.53913
$E[L]$	16.38302	16.22056

Table 7: Simulated and theortical variance

	Simulated variance	Theortical variance
$Var(L_{flat})$	17.36575	18.91064
$Var(L)$	18.71289	19.09773

In above the focus has been on putting a stress on the mortality rate, but a stress can also be applied on the spot rates. A relative decrease in the spot rates of $s\%$, the spot rates are decreased, the formula for the continuously compounded spot rate is $d(t) = e^{-t \cdot rt}$ which can be rewritten with an stress to following:

$$d(t)_{spot} = e^{-t \cdot (1-s/100)rt}$$

The variabel s is choosen so that $E[L_{trend}] = E[L_{spot}]$, as above the “trial and error”-metod is applied to find the value, and in this case $s = 8$. By using the new factor instead of the “original” in the simulation and theory, a new value in the variance and the expected value, which can be analyzed in “Table 8” and “Table 9”. The same pattern as with the deacrasing mortality rates, wich made an individual “live” longer, which implied to an higher expcted value, instead of an increasing life length, this stress makes future payemnts being more worth in todays money, and therefore making the expected value to increase. The variance in this scenario is the highest of all three types of stresses, and is therefore higher then $Var(L)$.

Table 8: Simulated and theortical expected value for total payments

	Simulated value	Theortical value
$E[L_{spot}]$	16.48011	16.53294
$E[L]$	16.38714	16.22056

Table 9: Simulated and theortical variance

	Simulated variance	Theortical variance
$Var(L_{spot})$	20.02381	20.22520
$Var(L)$	18.76432	19.09773

Summary

In “Table 10” all four expected values are listed, both the theoretical and the simulated values. It shows that $E[L_{trend}] = E[L_{flat}] = E[L_{spot}]$ and that all types of stress has made the expected value to increase. By having this in mind that a conclusion can be made that diffrent types of stress can produce the same expected value but the variance might be diffrent, which can be observed in “Table 11”.

Table 10: Simulated and theortical expected value for total payments

	$E[L]$	$E[L_{trend}]$	$E[L_{flat}]$	$E[L_{spot}]$
Simulated value	16.22056	16.54609	16.53913	16.53294
Theoretical value	16.16874	16.61539	16.45784	16.53365

Table 11: Simulated and theortical variance

	$Var(L)$	$Var(L_{trend})$	$Var(L_{flat})$	$Var(L_{spot})$
Theortical variance	19.09773	19.59565	18.91064	20.22520
Simulated variance	18.89928	18.54523	18.49275	19.44955

New estimations, confidence interval limits and comparisons

Based on the results from “Simulate 5 000 i.i.d. for a today 65 year old individual” we can now estimate the one year death probability and assign these values per year into a list. At the same presentation, a seen below, we compare it to the true ones from “life-table.txt”. The colons represent (Age), (Simulated death probabilities), (qx1000 from life-table.txt) respectively.

Table 12: Simulating and comparing death probabilities

Age	Sim P(Death)	qx1000
65	0.0040161	0.0059616
66	0.0054512	0.0064893
67	0.0065027	0.0070728
68	0.0073695	0.0077182
69	0.0082559	0.0084319
70	0.0079051	0.0092211
71	0.0102984	0.0100937
72	0.0101911	0.0110585
73	0.0111636	0.0121251
74	0.0121686	0.0133040
75	0.0161183	0.0146071
76	0.0193563	0.0160470
77	0.0167048	0.0176381
78	0.0215054	0.0193959
79	0.0185714	0.0213374

Age	Sim P(Death)	qx1000
80	0.0233918	0.0234815
81	0.0290873	0.0258488
82	0.0310238	0.0284620
83	0.0292709	0.0313457
84	0.0398450	0.0345270
85	0.0408986	0.0380356
86	0.0492596	0.0419037
87	0.0565134	0.0461665
88	0.0534813	0.0508623
89	0.0546293	0.0560325

The results, like from (i), comes from 5000 i.i.d. individuals.

The proceeding result shows the expected cash flow when simulated using the estimated death probabilities. One can conclude that the value are rather larger then the teoretical values which one would consider be due to the actual simulated values.

Simulated cash flow
16.41905

We proceed by simulating the variance with the use of the estimated death probabilities. We can observe that the result varies while tending to result near the theoretical value of the variance, as seen in the first assignment above.

Simulated variance
18.77834

Given the lower limits of the confidence interval, the following result was observed as the expected cash flow. By comparing with the given value of the expected cash flow when using the estimated death probability, one can conclude that the value given of the expected cash flow when using the lower limits of the confidence interval is slightly higher for each simulation. This since we then expect, according to the confidens interval, that fewer people pass away during the time interval and thus a higher value.

Simulated cash flow - Lower limits
16.61599

The following table compares the calculated Nelson-Aalen-estimator using estimations obtained from the simulations. This estimation is given by

$$\hat{A}(t) = \sum_{t_i: t_i < t} \frac{d_i}{y_i},$$

where d_i is the number of individuals that died at time t_i and y_i the number of individuals that were alive “just before” reaching time t_i .

Observed in the third colon is the *Gompertz-Makeham Law* and gives a type of result of death mortality. It is defined as

$$\alpha(t) = a + be^{ct}.$$

Since the function is an actual *rate*, one is forced to integrate it to be able to compare it with the Nelson-Aalen

estimator(NA Estimator). The information of the Makeham Law has been obtained from “Swedish M90”, which is the mortality rate investigation made by “Grundkommittén” in Sweden.

Table 16: Simulating and comparing death probabilities

Age	NA Estimator	Makeham
65	0.0074000	0.0059816
66	0.0056417	0.0065127
67	0.0087133	0.0071005
68	0.0077678	0.0077510
69	0.0067985	0.0084708
70	0.0085045	0.0092674
71	0.0098326	0.0101489
72	0.0105641	0.0111243
73	0.0111040	0.0122038
74	0.0140358	0.0133984
75	0.0166448	0.0147204
76	0.0182628	0.0161833
77	0.0163339	0.0178022
78	0.0209871	0.0195937
79	0.0223793	0.0215763
80	0.0231325	0.0237702
81	0.0259003	0.0261981
82	0.0270955	0.0288848
83	0.0278501	0.0318579
84	0.0313253	0.0351481
85	0.0348259	0.0387891
86	0.0469645	0.0428184
87	0.0402644	0.0472772
88	0.0513463	0.0522114
89	0.0514851	0.0576718

Further we now use the Nelsan-Aalen estimation once again and compare it with the given death probabilities of “life-tabke.txt”. Since the Nelson-Aalen estimator is simulated values, we see a slight difference in the mortality rates.

Table 17: Simulating and comparing death probabilities

Age	NA Estimator	qx1000
65	0.0060000	0.0059616
66	0.0080483	0.0064893
67	0.0070994	0.0070728
68	0.0091931	0.0077182
69	0.0086598	0.0084319
70	0.0072795	0.0092211
71	0.0092185	0.0100937
72	0.0103616	0.0110585
73	0.0121795	0.0121251
74	0.0136275	0.0133040
75	0.0146930	0.0146071
76	0.0140218	0.0160470
77	0.0187359	0.0176381
78	0.0200138	0.0193959
79	0.0206573	0.0213374
80	0.0246884	0.0234815
81	0.0245761	0.0258488
82	0.0350214	0.0284620
83	0.0308094	0.0313457
84	0.0363685	0.0345270
85	0.0380207	0.0380356
86	0.0380703	0.0419037
87	0.0477341	0.0461665
88	0.0482234	0.0508623
89	0.0606667	0.0560325

Based on a portfolio of size 500 of i.i.d. contracts, simulated 5 000 times

When simulating the expected value the insurance company will only make a profit 50% of the time, so to insure a profit margin they will add a safety loading to the premium: $\pi = E[L] + a$ where $a > 0$. In this task we will use $a = c \cdot \sqrt{\text{var}(L)}$ with c being to the 80th percentile and therefore $c = 0.84$ to secure profit. We have a portfolio of 500 respectively 5000 i.i.d contracts, therefore we can express the expected value of π as $E[\Pi_i] = E[L_i] + c \frac{\text{var}[L_i]}{\sqrt{\text{var}[L] \cdot n}}$ where n is the amount of contracts to insure.

Table 18: Total single premium values

Portfolio.Size.	Contracts 5000 (n)	Contracts 500(n)	Contracts 1(n)
500		16.38	19.79
5000	16.28		19.95